

Assignment 2 Case Study

Stage 2 Tutorial From Problems to Requirements

Week 5 | 60 minutes

Learning goals

- Analyse stakeholders.
- Distinguish functional and non-functional requirements.
- Recognise ambiguity and unsupported requirements.
- Define scope.
- Develop user stories and acceptance criteria.
- Critique AI-generated requirements.

Activity 1 - Stakeholder Map

Stakeholder	Need	Evidence
Management	A small, maintainable patient, practitioner, and appointment system	Needs low cost and for it to be simple, which may conflict with complex system features and tasks
Staff	To easily find patient information and avoid duplicate bookings	May prioritize speed and ease of use in terms of over detailed data entries, potentially resisting a complex new interface
Practitioners	Accurate appointment status and access to a reliable appointment history	Needs detailed clinical and historical data, which may conflict with management's goal of keeping the system simple and easy to use
Patients	Accurate booking information and consistent appointment statuses	May want self-service booking options, which might fall outside the current simple and easy to use system scope
IT Support	A small, manageable system that does not require difficult infrastructure or constant fixes because of bugs and other technical difficulties	May force back against staff or practitioners asking for advanced features in order to keep the system simple and easy to maintain

Activity 2 - Functional or Non-Functional?

- ☒ Functional ☐ Non-functional The system shall allow staff to cancel an appointment.
- ☐ Functional ☒ Non-functional The system should remain responsive for the course-scale dataset.
- ☒ Functional ☐ Non-functional The system shall retain cancelled appointments.
- ☐ Functional ☒ Non-functional Core business logic should be independently testable.
- ☒ Functional ☐ Non-functional The system shall search for a patient by ID.

Activity 3 - Repair Ambiguous Requirements

The system should be easy to use.

Problem: Easy to use is subjective and cannot be objectively tested/calculated. Clarification question: What is the maximum acceptable training time for new staff members to learn how to book an appointment?

Patient search should be fast.

Problem: Fast is a little bit vague and unclear, this means different things to different users. Clarification question: What is an acceptable response time for a patient info search to return results?

The system should securely manage data.

Problem: Securely is too vague as well and lacks specific technical standards. Clarification question: Which specific privacy laws and encryption standards must the system comply with when storing patient records?

Appointments should normally be easy to cancel.

Problem: Normally and easy provide no clear rules or guidelines. Clarification question: Under what specific conditions can an appointment be cancelled, and how easy should it be to book one?

Activity 4 - AI Requirements Audit

Classify each suggestion: Confirmed / Assumption requiring validation / Unsupported / Out of scope.

AI suggestion	Classification	Evidence / reason
Patients receive SMS reminders.	Unsupported	There is no mention/requirement of SMS or messaging needs in the client brief
Facial recognition login.	Out of scope	Management specifically requested a small and maintainable system
Receptionists create appointments.	Confirmed	The brief states staff deal with duplicate bookings, implying they are at fault and control of any misinput.
Online payment.	Unsupported	The brief limits the system scope to patient, practitioner, and appointment tracking/booking
Practitioners view schedules.	Confirmed	It is explicitly defined as a "practitioner and appointment system".
AI recommends treatments.	Out of scope	There is zero mention of clinical decision support or treatment features in the brief.
Cancelled appointments remain in history.	Confirmed	Staff currently struggle with limited appointment history, making historical tracking necessary.

Exit question

Why is 'AI suggested it' not sufficient evidence for a requirement?

AI generates suggestions based on generalized patterns and external data. Requirements must be rooted in direct evidence from the specific client brief, business goals, or stakeholder needs to ensure the final software solves the actual problem without introducing "out of scope" features.